

DONGDUCHEON, Korea, Republic ofWMO: **470980**Lat: **37.9019N**Lon: **127.0608E**Elev: **110**StdP: **100.01**Time Zone: **9.00 (E09)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-12.3	-23.5	0.5	-9.1	-21.4	0.6	-8.0	6.4	-3.5	5.8	-3.1	1.2	20	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	32.1	24.1	30.7	23.1	29.4	22.2	25.7	29.8	24.9	28.8	24.2	27.8	2.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	19.8	27.8	23.8	18.9	27.1	23.1	18.1	26.6	80.0	29.7	76.6	28.9	74.0	27.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.2	4.5	-16.9	34.3	2.6	1.5	-18.7	35.3	-20.3	36.2	-21.7	37.0	-23.6	38.1		
(5)			-17.6	26.8	2.4	0.8	-19.3	27.4	-20.7	27.9	-22.0	28.3	-23.7	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-3.5	-0.3	5.4	11.7	17.6	22.0	24.5	25.2	20.5	13.9	6.1	-1.6
	DBStd	10.55	4.26	4.14	3.93	3.56	2.75	2.11	2.19	2.46	2.54	3.48	4.43	4.41
	HDD10.0	1378	418	288	151	22	0	0	0	0	0	9	131	359
	HDD18.3	2983	676	521	402	201	47	2	0	0	10	141	366	617
	CDD10.0	2056	0	1	8	73	235	359	451	472	313	131	14	0
	CDD18.3	618	0	0	0	1	24	111	192	213	73	4	0	0
	CDH23.3	4591	0	0	0	26	255	771	1346	1772	401	21	0	0
	CDH26.7	1261	0	0	0	2	43	193	376	590	56	1	0	0
Wind	WSAvg	1.6	1.4	1.7	2.1	2.0	1.8	1.5	1.5	1.5	1.3	1.3	1.5	1.4
Precipitation	PrecAvg	1481	18	27	41	96	90	149	452	337	154	51	43	23
	PrecMax	2151	51	72	97	171	152	386	1102	901	489	152	112	68
	PrecMin	747	2	1	12	30	8	40	158	39	27	14	8	5
	PrecStd	347	14	20	24	37	41	81	237	235	118	33	27	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	14.0	19.9	26.2	29.9	31.7	33.4	34.4	29.6	25.4	19.8	11.5
		MCWB	5.6	7.7	10.0	14.4	17.2	19.8	24.9	25.5	20.9	16.9	13.4	7.0
	2%	DB	6.3	10.6	16.8	23.3	27.8	30.1	31.6	32.9	28.2	23.6	17.0	8.8
		MCWB	2.4	4.7	9.4	13.0	16.6	19.7	24.3	25.1	20.5	16.2	11.5	5.3
	5%	DB	4.5	8.4	14.5	21.0	26.2	28.9	30.1	31.4	27.0	21.8	15.0	6.9
		MCWB	1.1	3.6	7.9	11.4	15.8	19.5	23.7	24.4	19.8	15.1	10.4	3.6
	10%	DB	2.9	6.2	12.3	19.0	24.5	27.6	28.8	29.8	25.8	20.2	13.3	5.1
		MCWB	-0.1	2.1	6.4	10.6	15.3	19.2	23.1	23.7	19.0	14.2	9.5	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	10.4	12.6	16.5	19.6	22.9	26.4	27.2	24.1	19.6	15.1	8.0
		MCDB	8.4	12.4	16.6	22.7	24.7	27.3	30.6	32.2	26.8	21.7	17.9	10.3
	2%	WB	3.2	5.6	10.5	14.6	18.3	21.8	25.3	26.1	22.3	17.3	13.0	5.8
		MCDB	5.7	9.5	15.2	20.3	24.0	26.4	29.1	30.7	25.3	21.2	15.8	8.3
	5%	WB	1.7	3.8	8.7	13.1	17.3	21.1	24.7	25.3	21.3	16.2	11.3	4.0
		MCDB	3.8	7.7	13.2	18.1	23.3	26.0	28.4	29.5	24.7	20.4	14.5	6.5
	10%	WB	0.2	2.5	7.1	11.8	16.4	20.4	24.1	24.7	20.5	15.1	9.8	2.5
		MCDB	2.5	5.7	11.4	17.4	22.3	25.3	27.6	28.5	24.1	19.3	12.6	4.9
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.5	11.1	11.9	11.9	9.9	6.7	7.7	9.8	11.6	10.0	9.3
		MCDBR	10.5	12.7	14.1	15.4	15.3	12.9	9.5	9.7	11.4	12.8	11.7	10.2
		MCWBR	7.1	7.8	7.4	6.9	6.0	4.7	3.7	3.5	4.7	6.0	7.0	7.3
	5% WB	MCDBR	8.8	11.3	11.6	12.4	11.5	9.1	7.1	7.8	8.1	10.8	9.8	9.6
		MCWBR	6.5	7.8	7.1	6.8	5.5	4.0	3.2	3.4	4.1	5.7	6.8	7.4
Clear-Sky Solar Irradiance	taub		0.426	0.487	0.571	0.622	0.631	0.703	0.674	0.580	0.506	0.502	0.479	0.425
	taud		1.921	1.823	1.696	1.635	1.661	1.584	1.692	1.907	2.027	1.918	1.913	1.957
	Ebn at Noon		701	709	691	687	693	644	658	713	744	688	639	667
	Edh at Noon		141	178	222	250	247	267	238	188	158	159	142	128
All-Sky Solar Radiation	RadAvg		2.48	3.26	4.21	4.92	5.54	5.17	4.00	4.26	4.10	3.54	2.41	2.18
	RadStd		0.16	0.24	0.41	0.37	0.53	0.51	0.60	0.59	0.45	0.25	0.25	0.17